Final Course Project: AI, Economic Perceptions, and Consumer Sentiment Ahead of the 2026 Midterm Elections

Course: AI Applications in Finance

Duration: Six days

Deliverables: Research report, reproducible analysis, monitoring dashboard, and summary Power Point presentation

Project objective

Develop and validate an AI-assisted framework to measure economic narratives in news and social media and investigate whether they help explain differences between US consumer sentiment and measured economic conditions ahead of the 2026 midterm elections.

Your central task is to distinguish among three possible explanations:

- Consumers are responding to economic pressures inadequately captured by headline indicators.
- Consumers' perceptions reflect selective information, framing, or cognitive biases.
- The apparent discrepancy reflects measurement differences, timing, or an incomplete statistical model.

Do not assume that pessimism is irrational, that negative coverage is biased, or that media exposure causes declining sentiment. These are hypotheses to test.

Required research questions

Your project must address:

1. **How can we measure bias in news coverage or social media?**
2. **How can we determine the impact that such bias may be having on depressing US consumer sentiment, as measured by the University of Michigan survey, ahead of the upcoming midterm elections?**
3. How much of the apparent sentiment discrepancy can unemployment, inflation, and real GDP growth explain?
4. Which candidate perceptual mechanisms—such as negativity bias, availability bias, confirmation bias, or anchoring to past prices—can your data actually measure?
5. Does media analysis improve predictions of consumer sentiment beyond economic indicators alone?
6. What evidence would contradict a media-bias explanation?

1. Establish the economic and survey baseline

Construct a historical dataset with a monthly analysis frequency, preferably beginning in 2010, and a focused monitoring window covering the months preceding the 2026 elections. Include these required measures:

Measure	Required treatment
University of Michigan Index of Consumer Sentiment	Separate the headline index, Current Economic Conditions, and Consumer Expectations where available.
BLS U-3 unemployment rate	Use the seasonally adjusted official unemployment rate.

Headline PCE inflation	Calculate year-over-year inflation as 100 × (PCE price index / its value 12 months earlier − 1).
Real GDP growth	Use quarterly real GDP growth; clearly distinguish quarter-over-quarter annualized growth from year-over-year growth.

Michigan's survey covers personal finances, business conditions, and buying conditions, including expectations; it is not a direct rating of these three macroeconomic statistics. University of Michigan

U-3 measures unemployment relative to the civilian labor force. Obtain unemployment data from the BLS, and PCE prices and real GDP from the BEA PCE page and BEA GDP page.

Record observation periods, publication dates, and revisions. Use only information available by each prediction date. Do not interpolate quarterly GDP into artificial monthly observations or use subsequently released GDP estimates to explain earlier perceptions.

Estimate an economic-only model and define a **sentiment gap** as observed sentiment minus the model's out-of-sample prediction. Report prediction intervals. A negative residual is unexplained pessimism under your model—not proof of perceptual bias.

Test alternative explanations using a manageable set of controls, such as real household income, cumulative price increases, housing affordability, debt-service burdens, labor underutilization, or policy uncertainty. Explain why falling inflation need not reverse concern about already elevated prices, and why aggregate growth may differ from household experience.

2. Build a balanced media corpus

Collect economic coverage from a documented selection of national and local news sources and at least one social-media platform. If historical platform access is limited, narrow your claims accordingly.

Document:

- Search terms, dates, source selection, and inclusion criteria.
- Whether observations represent headlines, full articles, posts, or comments.
- Geographic relevance and available audience information.
- Duplicate stories, reposts, advertisements, and suspected automated activity.
- Missing data, access restrictions, and sampling limitations.

Distinguish **content supply**, **engagement**, and **actual exposure**. Likes and reposts are imperfect attention measures; they do not establish what survey respondents encountered.

3. Measure coverage and candidate biases with AI

Develop a coding scheme before applying an LLM or other NLP model. Measure at least four dimensions:

Dimension	Example operational measure

Tone	Negative, neutral, or positive economic language, separating factual description from emotional wording.
Selection and salience	Relative attention to layoffs, hiring, rising prices, falling inflation, growth, and recession risk.
Framing	Emphasis on price levels versus inflation rates, national aggregates versus household experiences, and current conditions versus forecasts.
Factual calibration	Whether numerical claims match the official information available at publication, with definitions and uncertainty handled correctly.
Political attribution	Which institutions or political actors receive credit or blame, without equating criticism with bias.

Define the benchmark against which each proposed bias is measured. Compare coverage of similar economic releases, equivalent positive and negative surprises, or the same events across sources.

Human-label a stratified sample of at least 300 items using two independent annotators. Report agreement, adjudication procedures, and AI performance by class and source. Compare the AI approach with a simple dictionary or conventional classifier.

Preserve model versions, prompts, and evidence excerpts. Test whether changing political names or source labels alters classifications of otherwise identical text.

Text can establish framing or selective coverage; it cannot, by itself, establish a reader's cognitive bias. Claims about confirmation or availability bias require corresponding exposure, recall, belief, or experimental evidence.

4. Evaluate association and causal impact separately

Compare three models:

1. Economic indicators only.
2. Economic indicators plus media measures.
3. Economic indicators, media measures, and additional household or political controls.

Use chronological holdouts or rolling forecasts. Report predictive accuracy, coefficient stability, uncertainty, and sensitivity to alternative specifications. Account for autocorrelation, seasonality, lag selection, and multiple testing.

For causal analysis, choose one feasible design:

- **Randomized framing experiment:** Assign consenting participants balanced factual summaries or alternative truthful framings of the same economic information. Measure changes in beliefs and sentiment-related responses.
- **Quasi-experiment:** Exploit plausibly external differences in media exposure, explaining treatment assignment, assumptions, spillovers, and pre-trends.
- **Observational analysis with a causal-design proposal:** Estimate associations and provide a detailed plan for stronger identification if suitable exposure data are unavailable.

Address reverse causality: pessimistic consumers may seek negative content, and publishers may respond to audience demand. Economic shocks may also affect both coverage and sentiment.

An experimental effect on your sample does not establish an effect on the national Michigan index. Explain what exposure estimates, representative sampling, weighting, and transportability assumptions would be needed to make that connection.

5. Track the pre-election period

Create an updateable dashboard displaying:

- Consumer sentiment and its economic-model prediction, with uncertainty.
- The sentiment gap.
- Media tone, framing, salience, and factual-calibration measures.
- Relevant economic releases and major events.
- Source composition, missingness, and model-performance indicators.

Where data permit, examine differences by income, age, geography, and political affiliation. Michigan has documented partisan differences in economic perceptions; treat affiliation as a potential moderator and confounder. University of Michigan

Assess survey-method comparability, including Michigan's transition to web collection and subsequent methodological reports. Michigan methodology announcement, 2026 reports

Do not attribute a change to election proximity merely because it occurred during the campaign. If your project finishes before election day, state the cutoff and provide a prospective update plan.

Deliverables and assessment

Submit:

- A **10-12-page report** answering the research questions and separating findings from hypotheses.
- A reproducible analytical package with data provenance, prompts, code, and a human-labeling guide.
- A monitoring dashboard.
- A 6-8**-minute presentation** explaining one practical implication for financial forecasting or risk management.

Criterion	Weight
Economic measurement, timing, and alternative explanations	20%
Corpus design and validated AI measurement	25%
Statistical analysis and causal reasoning	30%
Reproducibility and dashboard quality	15%
Communication, limitations, and research ethics	10%

Use public or appropriately licensed data, protect participant privacy, and obtain required institutional approval before human-subject research. Keep the project focused on measurement and financial analysis.

A rigorous finding that media bias does not explain the sentiment gap can earn full credit. The quality of the evidence and reasoning matters more than the direction of the conclusion.